

Bridging the Automation Gap

A Workforce Training Blueprint for ExFAB

A scalable 2-week intensive bootcamp model designed for 20–40 person cohorts, transitioning from macro discovery to an operational blueprint for the ExFAB biofoundry.

Industry Partner

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1.0 Executive Summary

The California NanoSystems Institute (CNSI) supports the ExFAB biofoundry, a high-throughput, AI-integrated research facility at UC Santa Barbara advancing automated biotechnology research. As ExFAB scales, its workforce challenge has become acute: the scientists and engineers who operate modern biofoundries do not yet exist in sufficient supply, and traditional academic training is not producing them.

This report presents findings from more than 60 semi-structured interviews conducted between January and May 2026 with professionals across biopharma, synthetic biology, agricultural technology, and clinical research. A structural gap exists between what universities teach and what automated laboratories require. New graduates arrive fluent in bench science but unfamiliar with workflow orchestration, LIMS environments, FAIR data principles, and cross-instrument troubleshooting. Industry respondents described this not as a knowledge deficit, but as a reasoning deficit: the inability to think in systems rather than procedures.

Three competencies emerged consistently across all stakeholder groups as the defining differentiators of effective biofoundry performance: the ability to troubleshoot autonomously across biological, hardware, and software domains; the capacity to translate scientific intent into machine-executable logic; and platform-agnostic adaptability, the ability to learn new systems rather than simply operate familiar ones. These competencies are not built through lecture-based coursework or vendor-specific certification. They require structured, hands-on exposure to real workflows, real failures, and real instruments.

In response, this report recommends a two-week, part-time bootcamp anchored in ExFAB's existing instrument suite and delivered through a blended dry-studio and wet-lab rotation model. Designed for cohorts of 20 to 40 participants, the program requires zero new full-time hires, no capital expenditure, and no facility modification, a deliberate constraint that preserves ExFAB's research-first mission while enabling scalable workforce impact. Financial projections indicate a net annual contribution of \$395,000 beginning in Year 2, with full recovery of the \$86,450 setup investment within the first cohort cycle.

2.0 The Workforce Skills Gap & Industry Findings

I. Misalignment Analysis

The current findings illustrate a structural misalignment between traditional university “manual-first” laboratory training and the digital, AI-integrated workflows that now define modern biofoundry operations. Academic programs continue to emphasize manual pipetting, single variable experimental design, and individual bench autonomy. Yet, automated research environments increasingly depend on competencies in workflow orchestration, data engineering, and system-level troubleshooting. Interviewees consistently identified this divergence as a primary barrier to scaling high-throughput experimentation. New graduates entering automated laboratories often encounter digital stress, ambiguity, and a sense of psychological contract violation as they confront unfamiliar automation logic, LIMS interfaces, and cross-instrument interdependencies. The limited academic exposure to simulator-based method development and multi-instrument reasoning compounds these challenges. These include competencies that ExFAB’s tier readiness matrices classify as essential for Tier 2–3 proficiency, particularly given the Echo 525’s lack of a functional simulator and the resulting requirement that every trainee above Tier 1 requires physical asset time.

The interviews reinforced how deeply this misalignment shapes early career performance. Academic training at UCSB and other institutions remains anchored in manual-first workflows. At the same time, industry increasingly depends on operators who can reason across multi-instrument systems, interpret automation logic, and articulate data-enhanced workflows. CNSI lab fellows described entering automated environments and ‘feeling like novices again,’ citing digital stress, ambiguity, and the abrupt shift from bench autonomy to orchestrating interdependent systems. Industry respondents echoed this gap, noting that new graduates “don’t know how to think in terms of workflows.” Thus, they struggle to translate biological intent into machine-executable steps and lack exposure to simulator-based method development. This is a particularly significant constraint for platforms such as the Echo 525, where hands-on time is the sole practical pathway for skill development.

Regional hiring constraints by California-based employers further emphasized that these constraints amplify misalignment, as companies cannot rely on a deep local talent pool with automation fluency, forcing them to invest heavily in retaining or recruiting from out of state. Collectively, the interviews demonstrate that the academic-to-industry transition is not hindered by scientific knowledge, but rather by the lack of structured preparation for digital and automated AI-augmented experimentation. This misalignment forms the

foundation for the broader workforce skills gap analyzed in subsequent sections, including implementation challenges, equipment bottlenecks, and hybrid competency frameworks that ultimately shape ExFAB's training and curriculum design.

II. Interview Synthesis

Across more than sixty industry interviews, employers consistently emphasized that the central constraint within automated biotechnology environments is the workforce's limited readiness to operate them. Respondents described a growing need for hybrid operators who can translate biological intent into machine-executable workflows, structure data in accordance with FAIR principles, and critically evaluate the outputs of AI-assisted tools. These capabilities require a blend of computational reasoning, experimental intuition, and cross-instrument situational awareness. Altogether, these competencies were repeatedly stated by employers as rarely cultivated within traditional academic training.

Additionally, interviewees highlighted a persistent implementation gap: companies can successfully pilot automation but struggle to operationalize it at scale. Their teams lack fluency in LIMS navigation, CSV-based protocol design, and the cross-functional communication required to coordinate biology, engineering, and data teams. Several employers consistently described early career hires as able to run an experiment but unable to run a workflow [Source: Recurring employer characterization — Theme 1, Interview Synthesis]. This reflects a clear distinction between executing a protocol and managing an automated system that integrates scheduling algorithms, error-handling logic, and data-capture pipelines. As one industry respondent put it, new graduates “don't know how to think in terms of workflows” [Source: Industry respondent, Interview Synthesis Section 4]. The operational fragility this creates was further illustrated by an AI lead at a synthetic biology startup, who noted that automated systems “require a lot of babysitting” and that tools that had been performing reliably could degrade unexpectedly after model updates, demanding constant human oversight rather than passive trust (Interview 20260422-002).

Moreover, sector-specific needs further sharpened these gaps. Agricultural technology employers emphasized the importance of high-throughput and phenotyping workflows, while biopharma respondents prioritized regulatory-aligned data integrity, synthetic biology startups stressed the need for operators who can rapidly iterate on design-build-test-learn cycles, and clinical research and diagnostics organizations emphasized documentation rigor, audit-trail integrity, and the ability to operate within validated, regulated workflows. Regardless of these differences, all sectors converged on the same theme: academic preparation does not yet produce workers who can reason across automated systems or troubleshoot at the interface of biology, hardware, and software.

Furthermore, California-based employers added a regional dimension, observing that the state's rapid growth within biomanufacturing has outpaced the availability of automation-ready talent. Companies described hiring pipelines constrained by a shallow local pool, forcing them to recruit from out of state or invest heavily in internal training programs. HR partners reported onboarding timelines of 45 to 90 days before new hires reach functional competency on site-specific automation workflows (Interview 20260505-001), reflecting both the complexity of the tools and the absence of structured academic pathways that prepare students for digital-first laboratory environments.

Collectively, the interviews reveal that the workforce challenge is not a lack of scientific knowledge, but a systemic gap in digital dexterity, workflow reasoning, and cross-instrument troubleshooting, skills that define modern automated experimentation but remain peripheral to university curricula.

III. Equipment Bottleneck

Industry interviews and ExFAB site observations intersect on a consistent set of bottleneck instruments. These tools impose the most stringent throughput constraints and require the deepest investment in training for safe and efficient operation, and they shape the training priorities for ExFAB's curriculum. Critically, the interviews make clear that the bottleneck is not equipment availability. The instruments exist; the constraint is workforce readiness and institutional implementation capacity.

This distinction emerged repeatedly across respondents. Employers did not describe shortages of hardware but rather the absence of structured pathways through which operators could develop the depth of understanding those instruments demand. As one automation lead at a mid-sized biopharma organization put it, skill-building today is fundamentally informal, a process of trial and error with no formal playbook and no repeatable entry point for new staff (Interview 20260428-001). An HR leader with experience supporting R&D organizations echoed this directly, noting that even strong candidates require months of onboarding before they can meaningfully operate automated systems, reflecting not the inaccessibility of the equipment but the absence of structured preparation that industry must compensate for on its own (Interview 20260508-001). The implication for ExFAB is precise: training infrastructure, not additional instrumentation, is the intervention the sector needs.

Within that framing, certain instruments concentrate the training burden more than others. The Echo 525 represents the most acute structural bottleneck in ExFAB's current portfolio. The complete absence of a first-party simulator means every trainee above Tier 1 requires physical asset time on the instrument itself, creating a hard constraint on cohort throughput that cannot be resolved through dry studio instruction. The

Tecan Fluent series introduces its own demands: hands-on calibration experience is essential, particularly when the instrument is integrated with scheduling software, and operators trained on a standalone configuration routinely misdiagnose failures at the workcell handshake boundary. Robotic arms and scheduling systems were cited more broadly as the most cognitively demanding training targets, requiring system-level reasoning capable of distinguishing hardware, software, and biological failure modes.

Sector-specific bottlenecks further sharpened these patterns. Agtech and plant science environments identified training gaps around quality control logic and error tracing across multi-step automated sequences (Interview 20260425-001). Biopharma and CDMO respondents concentrated on regulatory data integrity, with LIMS integration cited as the skill most consistently absent in early-career hires. Synthetic biology startups emphasized the ability to adapt protocols fluidly across rapid DBTL iterations without requiring expert intervention at each pass. In molecular diagnostics, the primary throughput constraints emerged at the LIMS interface layer, where manual workarounds for facilities lacking bidirectional integration created bottlenecks that automation was designed to eliminate (Interview 20260417-002). Food technology and industrial biotech respondents identified the discovery stage as the most consequential bottleneck, placing a premium on operators who could reason across both computational and physical experimental modes (Interview 20260215-001). Industrial biotech organizations additionally flagged operators trained on narrow, defined workflows who struggled to adapt when production processes diversified (Interview 20260227-001).

Collectively, these bottlenecks underscore the need for a structured, tiered training model that protects high-value assets while enabling progressive skill development. Across sectors, the pattern is consistent: instruments are available and in many cases underutilized, not because access is restricted but because the workforce pipeline does not yet produce operators prepared to use them effectively at scale. ExFAB's instrument master list and validated site visit inventory will anchor the final mapping of competencies to equipment tiers.

IV. A Validated Instrument List

ExFAB Instrument Portfolio 4 Categories · 8 Instruments

01 LIQUID HANDLERS	02 MASS SPECTROMETRY
Tecan Fluent 780 Piston / Air-Displacement	Thermo TSQ Altis Plus Triple Quad LC-MS
Beckman Echo 525 Acoustic Droplet Ejection	Thermo Orbitrap Ascend Tribrid High-Res LC-MS
	Thermo ISQ 7610 GC-MS
	Bruker neoflex MALDI TOF/TOF
03 CELL ANALYSIS & IMAGING	04 ROBOTICS & ORCHESTRATION
Agilent BioTek Cytation C10 Plate Reader / Imager	Thermo Spinnaker 3 SCARA Plate Robot
Beckman Coulter CytoFlex S Flow Cytometer	Thermo Momentum Workcell Scheduler

V. Hybrid Skills Taxonomy

EXFAB BIOFOUNDRY WORKFORCE TRAINING · UC SANTA BARBARA CNSI

Competency Framework — 5-Tier Progression

1 TIER 1 Initiator Identifies instruments, hazards, and safety protocols through written or verbal assessment. No hardware or software interaction required. GATE 1 Written knowledge test	SAFETY & IDENTIFICATION Instrument identification & purpose Hazard recognition (arms, chemicals, lasers, HV) PPE selection & safety checklist sign-off Emergency stop & interlock location FOUNDATIONAL CONCEPTS Platform architecture & arm / module functions SOP structure & lab access rules Instrument category context (LH · MS · CA · Robotics) ASSESSMENT Written quiz ≥80% + signed safety checklist — asynchronous, no instrument time required
2 TIER 2 Observer Shadows a Tier 3+ operator through a complete validated run. Loads labware from a deck map; reads back SOP steps as they execute. Uses software in read-only or offline mode. GATE 2 Simulator / offline software	OBSERVATION & SOP SKILLS Shadow a full validated method run Load labware from a written deck map SOP read-back while run executes Locate run status, error dialogs & simulation mode OFFLINE / SIMULATOR PRACTICE FluentControl 3D Simulator (liquid handlers) Xcalibur offline · Gen5 offline · Momentum Simulate Mode Protocol logic walkthrough without live hardware ASSESSMENT Supervised SOP walkthrough + observation log — sign-off by Tier 3+ operator; correct deck load without prompting

TIER 3 3 Operator

• BOOTCAMP EXIT

Independently loads, initializes, and runs validated methods. Interprets exception messages, executes error recovery, and performs routine operator maintenance — unassisted.

GATE 3 Physical demonstration on instrument

INDEPENDENT OPERATION

- Load labware, tips & initialize instrument
- Run validated methods end-to-end
- Interpret exception messages (LLD, tip-pickup, carrier errors)
- Execute standard error recovery & safe restart

MAINTENANCE & QC

- Daily System Care & operator-level PM
- Source cleaning · column install · MALDI sample spotting
- Daily QC runs · system suitability checks
- 5-error recovery on workcell (robotics)

ASSESSMENT Practical exam: run 2 validated methods unassisted; recover from 2 simulated errors — co-signed by Tier 5 ExFAB Fellow

TIER 4 4 Method Developer

Designs, validates, and documents new methods. Develops liquid classes, defines labware, integrates peripheral devices, and produces full validation packages. Mentors Tier 2–3 candidates.

ADVANCED Validation deliverable + Tier 5 review

METHOD DESIGN & VALIDATION

- Script / method authorship (modules, worklists, conditionals)
- Liquid class development with gravimetric validation ($CV \leq 5\%$)
- Labware definition & Zero-G / vision-guided teaching
- MS method development (DDA, PRM, SRM panels)
- Panel design · image analysis pipeline specification

INTEGRATION & DOCUMENTATION

- Peripheral device integration (RGA, Spinnaker, LIMS)
- IQ / OQ / PQ execution for new instrument additions
- SOP authorship & risk register
- Workcell nest teaching & motion database logging

ASSESSMENT Written method validation package (design + sim + validation + risk register) — signed by Tier 5

TIER 5 5 Super User / Trainer

ExFAB Fellows. Delivers Tier 1–3 cohort training, co-scores & signs all tier credentials, owns SOPs and vendor relationships, and holds driver-level troubleshooting authority. The zero-FTE delivery mechanism.

FELLOW ROLE Train-the-trainer + debug demo

TRAINING & CREDENTIALING

- Deliver full Tier 1–3 bootcamp training
- Co-score practical exams & sign tier credentials
- Gate 3 sign-off authority for cohort instrument access
- Challenge-out assessment for experienced entrants

TECHNICAL OWNERSHIP

- Driver / log-level debug (DriverFramework, Pegasus logs)
- Version-controlled SOP library & competency records
- Motion database “golden backup” & write-access control
- Calibration schedule & re-teach trigger management
- Tecan FluentSetup hardware certificate (liquid handlers)

VENDOR & COMPLIANCE

- Vendor escalation & FSE relationship (Thermo, Bruker, Beckman)
- USP <1058> Group C competency record maintenance
- Method-validation package review & approval

ASSESSMENT Train-the-trainer sign-off (deliver Tier 3 module to cohort, co-trainer scored) + driver-level troubleshooting demonstration

3.0 The 40–Person Blended Bootcamp Model

I. Capacity Scalability Strategy

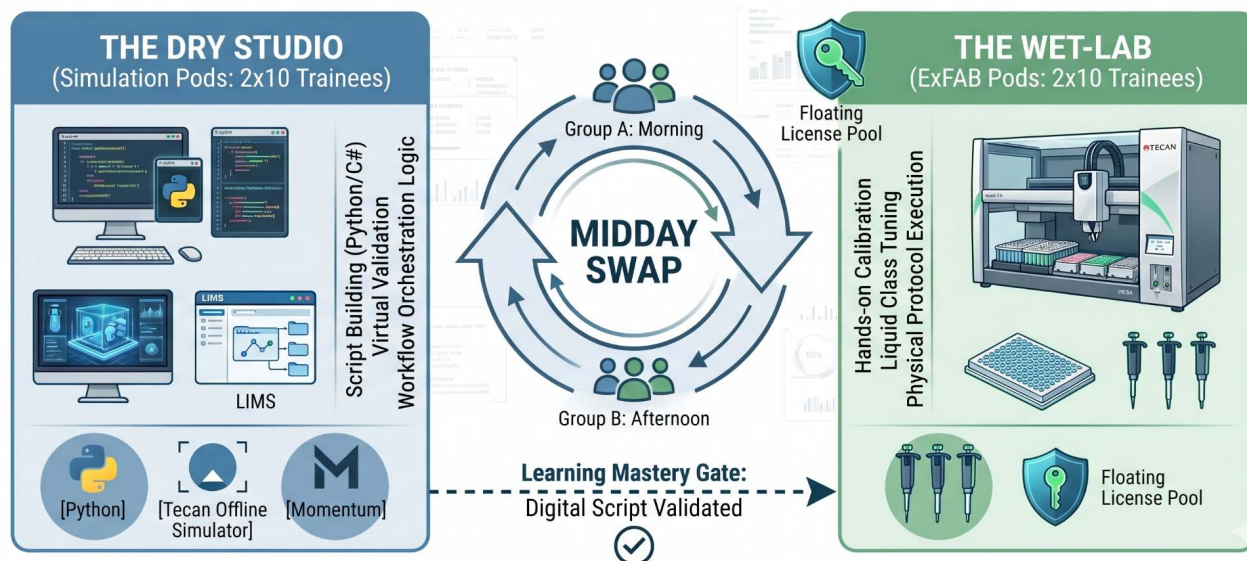
A core challenge for the ExFAB facility is accommodating 30 to 40 corporate trainees per cohort without disrupting the university's primary research activities. ExFAB prioritizes maintaining an 85–90% research-first oriented mission, which renders traditional high-volume physical training unscalable. To meet this capacity requirement with zero facility modifications, we propose a Blended Rotation approach that leverages digital twins, Python-based simulators, and no-code workflow canvases to shift the bulk of the educational burden away from physical equipment, dramatically expanding capacity while preserving ExFAB's core operations.

II. The “Blended Rotation” Logistics

The logistical architecture of the Blended Bootcamp is engineered around a parallel processing model to maximize facility utilization over a one-week intensive schedule. The 40-person cohort is divided into two distinct, alternating groups (Group A and Group B, n=20 each), allowing the program to process large numbers of trainees simultaneously without exceeding maximum room occupancy. While operating in the physical ExFAB space (the Wet Lab), trainees focus exclusively on high-stakes, tactile competencies that cannot be simulated. This includes hardware failure diagnosis, liquid class tuning, physical deck calibration, and the observation of pre-validated protocol runs. To ensure safety and adequate supervision, this group of 20 is further divided into micro-pods (e.g., a 10-person pod working with the Tecan 780, and a 10-person pod on the Beckman Echo 525). Operating concurrently in ExFAB's HPLC lab or in the classroom is the Dry Lab, where the alternate group utilizes floating software licenses to engage in digital simulation. Here, trainees map out complex LIMS data structures, write basic Python logic, and utilize digital twin software (e.g., Tecan FluentControl offline, Echo Plate Reformat) to design and virtually execute their protocols. At the midpoint of the day, a highly orchestrated logistical swap occurs. Trainees who spent the morning designing logic in the Dry Studio migrate to the Wet Lab to physically deploy their scripts. In contrast, the morning Wet Lab group transitions to the Dry Studio to analyze their physical run data and build their next iterations.

OPERATIONAL LOGISTICS: THE BLENDED ROTATION PLAYBOOK

Main Objective: Managing a 40-Person Cohort through Spatial Constraints (Dry Studio + Wet-Lab Swap)



III. Educational Design

Since the target demographic for this bootcamp comprises highly educated, incumbent bench scientists and working professionals, traditional pedagogical approaches (e.g., generic software tutorials or rote memorization) are insufficient. The curriculum is instead anchored in advanced adult learning principles (andragogy) and problem-based learning (PBL).

Biologists transitioning to automated systems frequently suffer from “black-box anxiety” – a deep-seated reluctance to relinquish experimental control to complex robotics. To dismantle this barrier, the curriculum employs progressive confidence-building. Instead of immediately exposing non-traditional programmers to pure, raw Python scripts, the Dry Studio initially introduces trainees to no-code, drag-and-drop workflow canvases, such as Synthace. Once the trainees understand the underlying logical mapping, they are gently transitioned into more advanced scripting environments.

Furthermore, the instructional design utilizes cognitive apprenticeships. Instead of merely teaching trainees how to execute a standard operating procedure (SOP), instructors focus on surfacing implied, algorithmic troubleshooting knowledge. Training scenarios are deliberately tied to real-world workflow anomalies – such as assay viscosity issues, dead-volume calculations, or robotic arm deadlocks – forcing trainees to think systematically and isolate biological failures from hardware failures.

IV. Instructional Model

A core strategic conclusion drawn by our consulting team is that for this program to be self-sustaining, it must operate without requiring any new full-time employees (FTEs). Rather than an external mandate, this “Zero FTE” constraint is a logical economic necessity. Hiring external, dedicated industry automation trainers is prohibitively expensive and would immediately render the bootcamp financially unviable for an academic core facility.

To address this operational bottleneck, our instructional model internalizes the teaching pipeline by leveraging ExFAB’s existing academic talent. Current ExFAB PhD fellows and post-doctoral researchers will be leveraged as paid “Tier 5 Super Users” to lead the training pods. This approach yields a dual-sided organizational benefit, as these fellows already possess an intimate, highly localized understanding of the unique spatial constraints, maintenance quirks, and operational realities of ExFAB’s specific hardware suite. By compensating these fellows with standard graduate student stipends or fellowship buyouts for the duration of the bootcamp, ExFAB dramatically reduces overhead costs while simultaneously empowering these young scientists with highly attractive, marketable industry training.

4.0 Financial Model

I. Mandate Compliance

A core conclusion drawn from our team’s insights is that the proposed bootcamp must operate within three design constraints that emerged consistently across interviews and were reinforced by CNSI leadership: zero new full-time employees, zero equipment purchases, and zero facility modifications. Our team approached this as the underlying financial logic of the bootcamp itself. This structure allows the bootcamp to scale workforce throughput without competing against ExFAB’s research-first operating mission.

To meet the zero-FTE constraint, our model uses the teaching pipeline by utilizing existing ExFAB PhD fellows and post-doctoral researchers as paid ‘Tier-5 Super Users’ rather than supplementing with external industry trainers. To meet the zero-CapEx constraint, our model expands trainee capacity by pairing digital-twin licenses with ExFAB’s existing instruments, eliminating the need to purchase additional physical hardware. To meet the zero facility modification constraint, the Blended Rotation Model is engineered to operate entirely within ExFAB’s current rooms, preserving the facility’s 85–90% research-first operating mission. Together, these constraints create a software-and-people investment that CNSI can absorb without disrupting its core mission.

II. Phase 1: Setup & One-Time CapEx

Phase 1 funds the one-time infrastructure required to launch the Dry Studio and onboard the first cohort. Our team has structured this investment to total \$86,450, distributed across four cost categories: purchasing hardware, floating software licenses, content development, and industry outreach.

Hardware is the largest line item at \$25,000, representing 10 CAD-level engineering workstations with dual monitors at standard enterprise pricing (approximately \$2,500 per workstation from vendors such as Dell or Lenovo). These workstations in the Dry Studio serve as the physical hosts for the floating-license simulator, providing trainees with digital twins of the physical instruments in the Wet Lab.

Floating software licenses total \$36,450, distributed across three vendor platforms (Beckman Coulter Echo Plate Reformat at \$13,950, the Tecan offline simulator suite at \$10,500, and Thermo Momentum at \$12,000). Our team recommends licensing each platform at three concurrent seats to support paired-station training across the cohort, with per-user licensing. Per-seat pricing was benchmarked against public academic records and comparable pro-formas for offline programming.

Content development accounts for building Tier 1–3 curriculum modules, simulator walkthroughs, and mastery assessments within an existing learning-management system. Our design structures are built around progressive mastery gates that require trainees to validate digital simulation logic before being granted access to physical instruments, thereby protecting high-value assets from premature operation. One to two students can develop this internally, eliminating the overhead cost of an external SaaS contract. Marketing and industry outreach are budgeted at \$5,000, leveraging the pipeline of more than 400 industry leads our team sourced during the customer-discovery phase.

III. Phase 2: Annual Operating Expenses

Phase 2 presents the recurring annual cost of running one cohort alongside internal UCSB throughput. Our financial model projects the total annual operating expense at \$115,000, a figure that's lower than what an equivalent equipment-heavy bootcamp would require because the Blended Rotation Model is engineered to shift initial trainee failure and material waste away from the wet lab and into the digital twins.

Personnel costs total \$45,000 for teaching-assistant and graduate-student support, compensating ExFAB fellows leading bootcamp rotations under our 'Tier-5 Super User' model. Drawing on existing graduate and post-doctoral researchers as the instructional layer, the program avoids the cost of new full-time hires while equipping these fellows with marketable industry training experience. Facilities access fees total \$20,000,

representing an internal chargeback to CNSI cores at negotiated off-peak rates, down from a \$50,000 baseline.

Materials and consumables are budgeted at \$15,000, reflecting the operational reality that trainees no longer consume physical materials on failed runs; initial-failure waste is absorbed inside the simulators, with consumables drawn only for digitally validated protocols. Ongoing software licensing accounts for \$20,000 for LMS maintenance and vendor-simulator integration. Digital-twin and hardware maintenance accounts for \$15,000, modeled at a standard 15–20% annual maintenance rate for enterprise software and hardware (Podmurnyi).

IV. Phase 3: Annual Revenue

Our financial model projects annual revenue of \$510,000, generated across four streams that buffer the program against reliance on any single funding source.

Grant funding accounts for \$230,000, targeting federal workforce-development vehicles aligned with ExFAB's Tier 4 hybrid-competency mandate. These include the National Science Foundation STEM Workforce Development program, the CHIPS Act workforce allocations, and the NIIMBL training-grant pipeline. Our team has set the \$230,000 target conservatively within the typical \$200,000–\$500,000 range for NSF workforce development awards.

Corporate training fees amount to \$200,000, modeled on the NIBRT pricing structure of one cohort per year, with 40 trainees at \$5,000 per seat (NIBRT, 2025). Our team has positioned the \$5,000 seat price above NIBRT's published baseline because the proposed curriculum delivers a hybrid digital-twin component that NIBRT's GMP-focused program does not offer, extending the same simulator-led pedagogy into an upstream R&D context.

Industry partnerships account for \$50,000, structured as two Gold-tier corporate sponsorships of \$25,000 each. Target sponsors include equipment vendors and California-region startup employers, and in exchange, sponsors receive early access to our validated 'Tier-5' trainer pipeline. Candidate sponsors in both segments have already been identified through our outreach. Individual student fees and certifications account for the remaining \$30,000, drawn from institutional offsets for UCSB students completing the bootcamp. This stream aligns with the Association of Biomolecular Resource Facilities (ABRF) micro-credentialing framework, which is increasingly adopted across academic core facilities.

\$86,450 One-Time Setup Cost	\$115,000 Annual Operating Expenses	\$510,000 Annual Revenue	\$395,000 Net Annual Profit (Year 2+)
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Phase 1: One-Time Setup (CapEx)	
CAD-Level Workstations (10)	\$25,000
Echo Plate Reformat (3 licenses)	\$13,950
Tecan Offline Simulator (3 licenses)	\$10,500
Thermo Momentum (3 licenses)	\$12,000
LMS Setup	\$20,000
Marketing & Industry Outreach	\$5,000
Total	\$86,450

Phase 2: Annual Operating Expenses	
TA / Student Support	\$45,000
Equipment Access Fees	\$20,000
Consumables	\$15,000
Software Licenses (LMS)	\$20,000
Digital Twin / Hardware Maintenance	\$15,000
Total	\$115,000

Phase 3: Annual Revenue	
NSF / CHIPS Act Grants	\$230,000
Corporate Training Fees	\$200,000
Industry Partnerships	\$50,000
Student Fees / Certifications	\$30,000
Total	\$510,000

Annual Revenue	Annual Operating Expenses	Net Annual Profit (Year 2+)	Year 1 Net (after setup)
\$510,000	(\$115,000)	\$395,000	\$308,550

V. The NIBRT Model

A central question raised during our team’s financial modeling was whether a simulator-led training program operating at this scale could realistically generate revenue for CNSI. We anchored that question to a published institutional case: the National Institute for Bioprocessing Research and Training (NIBRT) in Ireland, which operates as a global hub for simulator-led GMP training and has demonstrated a sustained positive return on investment for its industry partners (NIBRT, 2025).

Our proposed Biofoundry-as-a-Service model is engineered to adapt the NIBRT operational structure; accurate simulators, short cohort cycles, and a corporate-fee revenue base, to a U.S. academic core-facility context. The critical adaptation is that our model preserves the underlying NIBRT economics while honoring CNSI’s zero-FTE and research-first constraints.

VI. Organizational ROI

With a \$86,450 one-time setup cost, our financial model projects \$510,000 in annual revenue against \$115,000 in annual operating expense, producing a net annual contribution of \$395,000 that recurs from Year 2 onward. Year 1 itself remains net-positive at \$308,550, meaning the program is structured to cover its own setup within the first cohort cycle.

Our team’s central financial argument is that the proposed bootcamp directly resolves the “Build vs. Buy” dilemma that surfaced repeatedly across our sixty-plus industry interviews. Our research indicates that the central workforce constraint is the absence of hybrid operators who can translate biological intent into

machine-executable workflows. External hires with this profile are scarce and expensive, and for most small-to-mid-sized organizations, the internal upskilling of existing staff is the more logical and strategic option. The proposed bootcamp operationalizes that upskilling at a \$5,000-per-seat price point is a fraction of the replacement-cost penalty the industry incurs when it loses trained automation talent. Digital-twins further reduce consumable waste and instrument downtime, an asset-protection benefit that is well-documented as a primary ROI driver of digital-twin adoption in industrial settings (Podmurnyi).

5.0 Workforce Training Recommendations Matrices

I. What This Section Delivers

This section converts the skills-gap findings of Sections II–III into three tools ExFAB can put to use without hiring, purchasing, or modifying anything. The first is an Actionable Mapping that ties every workforce gap we identified to a specific ExFAB program element. The second is a set of instrument-specific competency benchmarks that state, in plain operational terms, what a trainee can do at the target exit tier and what an ExFAB Fellow must own as a Super-User. The third is a Micro-Credentialing & Facility Access System that controls who is allowed to touch which instrument, and when. Every recommendation here is designed to operate inside Sherylle’s three non-negotiables: 30–40-person cohorts, zero new FTEs, and no facility modifications, and a deliberate focus on the high-demand “stickler” instruments rather than the entire portfolio.

II. The Actionable Mapping: From Industry Gaps to ExFAB Offerings

The table below is the primary deliverable. Each row is a workforce gap surfaced repeatedly in our industry interviews; the right-hand side of each row is the specific ExFAB program element that addresses it.

Industry Workforce Gaps & ExFAB Program Response

#	WORKFORCE GAP	EXFAB OFFERING THAT ADDRESSES IT
1	New hires lack a fundamental understanding of how automated equipment works — independent of scientific or biological background.	Instrument-agnostic Tier 1-2 module delivered via Gate 1 written test and Gate 2 simulator.
2	Troubleshooting capability separates effective from ineffective operators and is the hardest skill to source.	Tier 3 Operator competency builds independent error-recovery into every instrument matrix, validated at Gate 3 physical demonstration.
3	Automation skill is built informally and unrepeatably — "forged by fire," with no formal playbook.	Structured 1–2 week blended bootcamp plus four instrument skills matrices converting tacit experience into observable, assessable behaviors.
4	Operators cannot reliably distinguish biological vs. hardware vs. software failure.	Tier 4-5 failure-triage competencies across all matrices: DriverFramework/log-level diagnosis (liquid handlers, robotics); result-vs-instrument QC logic (mass spec, cell analysis).
5	Vendor lock-in: operators trained on one platform cannot transfer skills to another.	Vendor-agnostic Tier 1-3 HTE logic core taught on simulator before platform-specific software layers, with explicit instruction on the platform divide.
6	Wet-lab biology fundamentals gap among engineering-background hires (pipetting, PPE, decontamination).	Bootcamp foundational module plus Tier 1-2 safety and SOP-observation requirements embedded in every instrument matrix.
7	LIMS/instrument-to-LIMS integration is a scarce, high-value skill.	Tier 4 Method-Developer competencies (structured data export, worklist, and LIMS handoff) — positioned as an advanced elective, not core bootcamp content.
8	Training cannot scale to 30–40 people without new FTEs, redundant equipment, or facility modifications.	Gate 2 simulator rotation, ExFAB Fellows as Tier 5 Super-Users (zero-FTE delivery), and stackable per-instrument credentials to ration scarce physical instrument time.

Two interpretive notes matter for leadership planning.

Note 1: In-house vs. vendor touchpoints. Nearly all offerings run on existing ExFAB equipment, Fellows, and freely deployable software (Xcalibur 4.7+, FreeStyle, Restek Pro EZGC, Gen5, CellProfiler). Only two vendor engagements are required: (a) Tecan FluentSetup hardware certification for the Tier 5 liquid handler lead, plus a commercial inquiry on multi-seat FluentControl simulation licensing; and (b) the Bruker Installation Familiarization event for the neoflex MALDI, with optional Essential Operator courses. All other training is in-house. Limiting vendor dependency to two known line items is what makes the zero-CapEx posture defensible.

Note 2: Bottleneck instruments vs. foundational skills. Physical instrument time is deliberately concentrated on high-demand platforms: Tier 3 liquid-handler competency and robotics-orchestration tiers directly target the Tecan/Spinnaker workcell throughput chokepoint. Foundational skills, equipment literacy, safety, SOP read-back, are taught once at Tiers 1–2, instrument-agnostically, and transfer across the full portfolio. This separation is what makes a 30–40-person cohort tractable: instrument time is rationed to competencies that genuinely require it.

This structure responds directly to the industry’s description of the current state. As one automation engineer running a 30-plus-instrument Tecan fleet put it, today’s skill-building is trial and error: you get forged by fire, with no formal playbook; it really is accumulated experience over time (Interview 20260428-001). The Actionable Mapping is, in effect, the playbook that does not currently exist.

III. The Competency Benchmarks: What Each Tier Looks Like at ExFAB

Rather than reproduce four instrument deep-dives in the body of this report, the table below gives one row per instrument family. It shows only the two tiers leadership needs to reason about: the Tier 3 exit standard (where most bootcamp trainees finish) and the Tier 5 Super-User responsibility (what an ExFAB Fellow must own for the model to run without new staff).

Instrument Competency Standards — Tier 3 & Tier 5

INSTRUMENT FAMILY	TIER 3 — OPERATOR	TIER 5 — SUPER-USER
Liquid Handlers Tecan Fluent 780 Beckman Echo 525	– Loads labware independently; runs validated methods; interprets common exceptions (LLD errors, tip-pickup, carrier misalignment) and executes standard recovery. – Performs operator-level maintenance; ≥4 hrs physical instrument time required.	– Delivers Tier 1-3 cohort training; co-scores exams and signs credentials; debugs at DriverFramework / Pegasus log level. – Liaises with Tecan support; holds or is actively pursuing Tecan FluentSetup hardware certificate.
Mass Spectrometry TSQ Altis Plus Orbitrap Ascend ISQ 7610 GCMS Bruker neoflex MALDI	– Completes validated method end-to-end with QC review; operator-level maintenance (source cleaning, column install, MALDI sample spotting). – Passes CLSI QMS03 practical exam and blind QC accuracy check; 8-16 hrs instrument time (neoflex at high end).	– Reviews and approves method-validation packages; maintains version-controlled SOP library; maintains competency records per USP <1058> Group C. – Owns Bruker FSE relationship and Essential Operator course bookings.
Cell Analysis & Imaging Agilent BioTek Cytation C10 Beckman Coulter CytoFlex S	– Independent tube-mode acquisition of validated 1–2 color microbial reporter panel; plate-reader and basic widefield imaging with daily QC. – ~4-8 hrs supervised instrument time; plate-loader mode is a separate advanced competency.	– Owns Gen5 / CytExpert protocol libraries and QC standards; trains and signs off cohorts; owns vendor escalation. – Gatekeeps advanced plate-loader and image-analysis competencies.
Robotics & Orchestration Thermo Spinnaker 3 Thermo Momentum	– Loads validated Momentum protocol; monitors scheduler; pauses/resumes safely; recovers independently from five common errors (plate drop, missed pickup, vector collision, hotel/stacker error, home-position loss after E-stop). – ~4-8 hrs supervised robot time required.	– Owns motion database and restorable golden backup; countersigns all Tier 4 teach logs and re-teach validations. – Owns calibration schedule and re-teach triggers; controls write access to motion database.

The key structural insight: the liquid-handler divide. The single most important distinction in ExFAB's portfolio is not between instrument categories but within the liquid handlers — between the piston-based Tecan Fluent 780 (FluentControl) and the acoustic-droplet-ejection Beckman Echo 525 (Echo Cloud). (An earlier planning draft framed this as a “Hamilton vs. Tecan” comparison; that was a carry-over error — ExFAB's actual liquid handlers are the Tecan Fluent 780 and the Beckman Echo 525, and the benchmarks above are corrected accordingly.) This divide matters because Tier 3 on a liquid handler is the highest-ROI training target in the entire program: the underlying logic of high-throughput experimentation — deck layout, labware definition, run staging, exception handling — partially transfers across platforms, so the training investment pays back on more than one instrument. By contrast, instrument-specific mastery, such as MALDI crystallization technique on the neoflex or multi-color compensation on the CytoFlex, does not transfer; it is an advanced elective territory, appropriate for returning users rather than the core cohort.

Full instrument-by-instrument tier descriptors, with assessment criteria for all five tiers, are found in the **Skills Matrices** Folder.

One technical gap leadership must know about. The Bruker neoflex MALDI has no offline simulator — FlexControl requires a live hardware connection, and there is no instrument-level digital twin. This makes the neoflex the highest physical-access-time instrument in the portfolio (8–16 hours of instrument time for the Tier 2→3 transition, with minimal reduction available from software workarounds because crystallization optimization is inherently physical). Practically, it is the one instrument whose cohort scheduling must be coordinated most tightly against ExFAB's 85–90% research mission, and it should be planned for significantly more rotation time per trainee than the other three families.

IV. The Micro-Credentialing & Facility Access System

This is the most operationally consequential — and the most debatable — recommendation in the section, so it is grounded directly in what industry practitioners told us they already do. The system is three sequential gates. A trainee cannot reach a gate until the previous one is cleared.

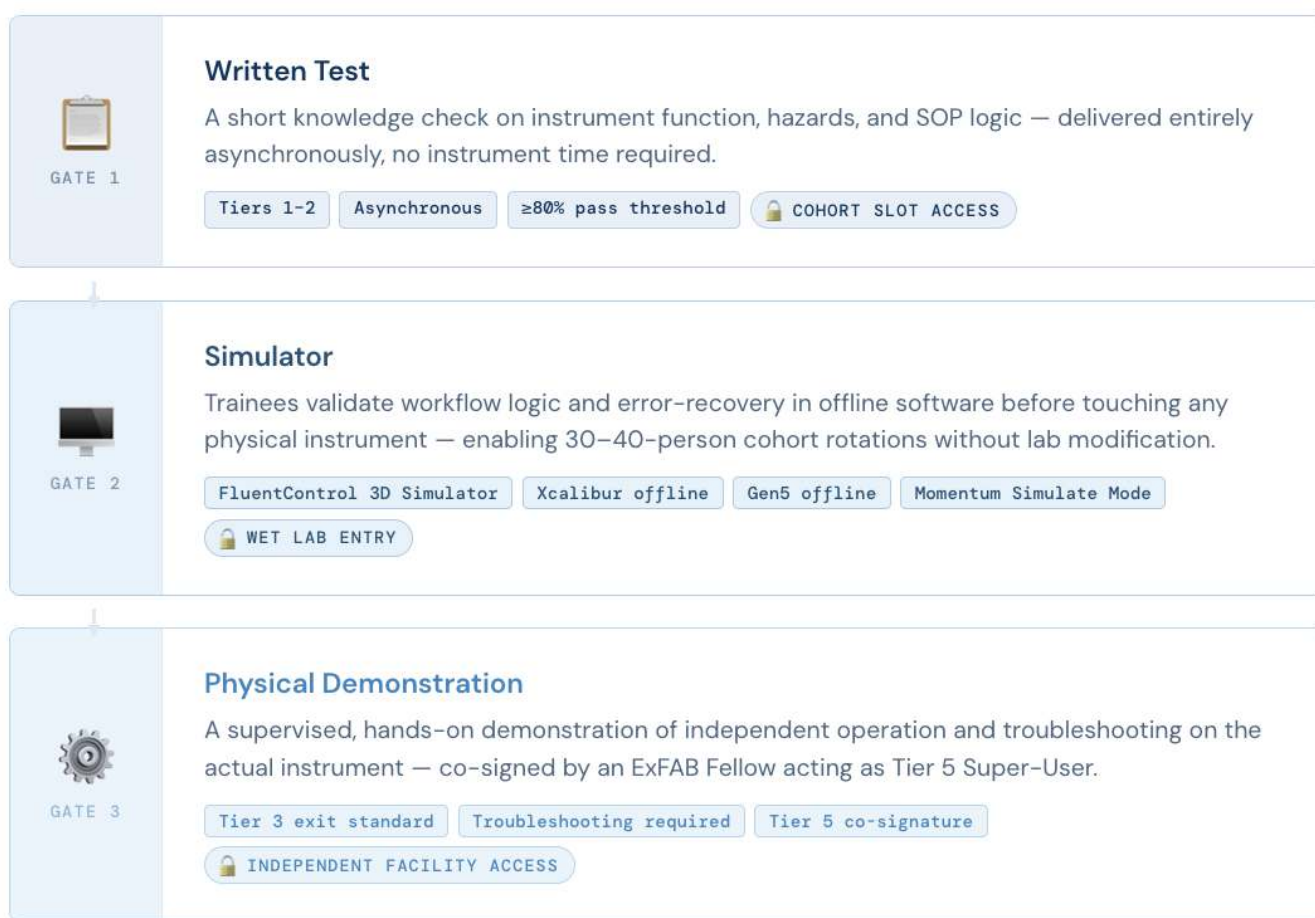
1. **Gate 1: Written test.** A short knowledge check on instrument function, hazards, and SOP logic. Its purpose is administrative triage: it keeps unprepared users out of scarce cohort slots before they consume any instructor or instrument time. It can be delivered and scored entirely asynchronously.
2. **Gate 2: Simulator.** Trainees validate workflow logic in offline software (FluentControl 3D Simulator, Xcalibur offline, Gen5 offline, Momentum Simulate Mode) before touching any instrument. This is the gate that directly solves the physical-space constraint Sherylle raised: it lets ExFAB run 30–40 people through logic and error-recovery training in a dry studio while the wet lab continues its research mission, with the two groups rotating. It is the mechanism that makes the cohort-size mandate achievable without modifying a single labroom.

3. **Gate 3: Physical demonstration.** A supervised, hands-on demonstration of competence — including troubleshooting — on the actual instrument. This is the real gatekeeper before independent facility access, and it is co-signed by an ExFAB Fellow acting as the Tier 5 Super-User. No new full-time staff is required because the sign-off authority resides with the personnel ExFAB already has.

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Three-Gate Credentialing System — Sequential Progression

A trainee cannot reach a gate until the previous one is cleared. Each credential is scoped to the instrument, not the facility.



This three-gate architecture is not a theoretical construct — it mirrors what industry already practices. The day-one gap that matters most, according to a clinical-lab automation manager, is not biology pedigree: it doesn't matter the ... scientific background or biological background that you have; what the same interviewee identified as the trait that separates their best automation employee was that their

troubleshooting capability really stands out (Interview 20260413-001). That is precisely why troubleshooting is the explicit pass condition at Gate 3 rather than an afterthought.

The three-gate, signed-off progression also matches established industry practice almost exactly. One laboratory supervisor, describing their own onboarding without being prompted toward a tiered model, explained: we don't throw anyone off the deep end, and that for facility access, troubleshooting is one of our check boxes that they have to understand — trainees must get trained on it, get signed off, and then they have a trainer who signs off saying, yes, this person is adequately trained (Interview 20260417-002). This is the independently arrived at Gate 3 design.

The stackable credential architecture. Credentials are scoped to the instrument, not the facility. Reaching Tier 3 on a liquid handler does *not* grant access to the neoflex MALDI or the CytoFlex. Each instrument family is its own gated credential, and a returning user re-enters only at the gate appropriate to the new instrument. This protects expensive equipment from users credentialed on something else, and it gives returning industry participants a clear, additive progression path rather than an all-or-nothing certification.

V. Implementation Considerations

Three points require a leadership decision before any bootcamp launches.

1. **The ExFAB Fellows Super-User pathway must be confirmed first.** The Fellows are not a supporting element of the delivery model — they *are* the delivery model. Every Gate 3 sign-off, every Tier 1–3 training cohort, and every vendor-escalation relationship in Section C depends on Fellows operating at Tier 5. If that pathway is not confirmed and resourced, nothing downstream in this section is executable.
2. **One vendor prerequisite is non-negotiable regardless of cost.** Whoever holds Tier 5 on the liquid handlers must obtain the **Tecan FluentSetup hardware training certificate**. It is the single credential in the entire program that cannot be satisfied by bootcamp training and requires direct engagement with Tecan. It should be treated as a fixed prerequisite rather than a discretionary line item.
3. **Mastery-based progression needs an accelerated track for experienced entrants.** Rigid gatekeeping without flexibility risks alienating exactly the senior-level participants ExFAB most wants to attract. Industry flagged this directly: a mastery model has to be ... fluid, because if you bring someone ... that really ... has experience, then I think it could be a delay, but if the structure is flexible enough so that two people who can master things quickly are able to move through ... then yeah, I think that works (Interview 20260413-001). The recommendation is therefore not to relax the gates

but to allow accelerated challenge-out: an experienced entrant who can demonstrate Gate 3 competence is not required to sit through Gates 1–2 in real time. This preserves asset protection while removing the “waste of time” objection that would otherwise drive away the program’s most valuable audience.

6.0 Career Pathways

I. Emerging Hybrid Roles

As automated experimentation becomes the operational backbone of modern biofoundries, the workforce is shifting towards a new class of hybrid scientific roles. These positions merge biological intuition with computational reasoning, workflow architecture, and system-level troubleshooting. Across interviews, employers repeatedly emphasized the importance of competencies required to run high-throughput and AI-augmented laboratories, which traditional job titles no longer implement. Rather, organizations are hiring for roles relating to automation scientists, biofoundry operations, DBTL (Design-Build-Test-Learn) engineers, workflow architects, and data-fluent biologists, all of whom sit at the intersection of biology, engineering, and informatics (Delebecque & Philp, 2019; Mey et al., 2021).

As described by Interviewee 20260313-001, these roles entail “one foot in biology, one foot in hardware and software,” requiring individuals who can translate scientific intent into machine executable logic, structure data according to FAIR principles, and diagnose failures across biological, mechanical, and computational domains. Employers also noted that the most effective team members are those who “aren’t afraid to crash the machines,” as stated by Interviewee 20260309-001, who can reason through multi-instrument dependencies and who can communicate across cross-functional teams (Stephenson et al., 2023). These hybrid roles are rapidly becoming the norm across biopharma, ag. tech, clinical diagnostics, synthetic biology, environmental biotech, and industrial biomanufacturing.

For CNSI and ExFAB, this shift represents a strategic opportunity, as the facility is uniquely positioned to prepare students for these emerging roles. Industry leaders consistently identify these competencies as missing from the academic pipeline, and by teaching universal automation logic, FAIR-aligned data structuring, and cross-platform workflow design, they can get ahead on the opportunity curve. Instead of training students on a single vendor’s interface, ExFAB can cultivate a workforce capable of navigating diverse automation ecosystems and adapting to rapidly evolving tools. This approach supports long-term

operational flexibility and aligns with CNSI's mission to build open and scalable scientific infrastructure that serves both academic and industry partners.

II. Vendor-Agnostic leadership

A recurring theme across interviews consisted of the industry's growing concern about over-specialization and vendor lock-in. Employers described teams that could operate a single instrument but struggled to generalize their skills across platforms, had limited organizational agility, and slowed the adoption of new tools. Several automation leads noted that durable competencies are not tied to any specific interface, but rather to the underlying operational logic. This includes how liquid classes behave, how deck layouts constrain movement, how scheduling dependencies shape run feasibility, how errors propagate, and how data provenance must be maintained.

ExFAB is well-positioned to address this directly. Rather than training students on any single platform, the curriculum is built around the underlying logic that transfers across systems. This means understanding how liquid classes are defined and why factors such as aspiration speed, backlash volumes, and surface tension affect transfer accuracy. It means knowing how deck layouts constrain movement, how scheduling dependencies determine run feasibility, and how data provenance must be maintained throughout a run. These are the competencies that hold regardless of whether a lab runs Hamilton or Tecan, Momentum or Green Button Go. One interviewee put it plainly: teams tend to build momentum around specific vendors and stay in their bubble, and when those tools change, or a new platform gets brought in, that institutional knowledge doesn't transfer. By grounding students in vendor-agnostic principles from the start, ExFAB can produce genuinely adaptable graduates, not just trained on the instruments available during their rotation.

III. Implementation Roadmap for Career Pathway Integration

To translate these insights into an actionable workforce development, ExFAB should pursue a phased implementation strategy that aligns with operational realities and leverages existing fellow expertise.

The first priority is to validate the blended rotation model with current ExFAB fellows. Fellows consistently emphasized that hands-on exposure, paired with simulator-based reasoning, accelerates learning and reduces digital stress for new users, and that a blended rotation structure that combines wet and dry labs mirrors how industry teams onboard new hires. Alongside this, ExFAB should finalize the tier 3-5 equipment matrix. Interviews identified Hamilton, Tecan, Echo, and scheduling systems such as Momentum and Green Button Go as the highest-demand and most complex tools, and mapping these to ExFAB's tiered readiness framework will clarify training pathways and protect instrument runtime.

Before launching the pilot, ExFAB should also conduct a final round of industry outreach to pressure-test the curriculum, validating it with automation leads from LanzaTech, Genomatica, agtech partners, and clinical labs to ensure alignment with real hiring needs. This outreach runs in parallel with preparing the pilot bootcamp for the summer/fall cohort, which includes developing simulator modules, drafting vendor-agnostic SOPs, and coordinating with fellows who will serve as Tier 5 super user instructors.

Once the pilot cohort is complete, ExFAB should use outcomes data to refine and scale the model. Fellows who demonstrate advanced proficiency can transition into a post-cohort residency track, taking on cross-instrument integration projects under Tier 5 supervision. Over time, this creates a self-sustaining pipeline in which today's trainees become tomorrow's super-user instructors, reducing the program's dependence on any single cohort of fellows. As the program matures, ExFAB is also well-positioned to document and publish its pedagogy outcomes, contributing first-of-kind training benchmarks to the broader biofoundry field. Altogether, these steps position ExFAB to become a regional leader in preparing students for the hybrid scientific roles that now define automated biotechnology.

7.0 Conclusion

I. Synthesis of Strategic Shift

This project marks a strategic transition from identifying macro-level workforce challenges in biotechnology to delivering an actionable operating blueprint for the ExFAB facility. At the midpoint stage, our research identified several major problems shaping the field of automated biotechnology, including curriculum lag, the build-versus-buy dilemma, vendor-specific training limitations, and the shortage of hybrid professionals who can bridge biological expertise with automation logic, data literacy, and system-level troubleshooting. The final report moves beyond this discovery phase by translating those findings into a practical training model tailored to ExFAB's facility, instruments, staffing structure, and research-first mission.

The core conclusion is that ExFAB's challenge is not only about access to automation technology but also about building a scalable training system around it. As automated workflows become more central to biotechnology R&D, CNSI needs a way to convert expert knowledge, equipment access, and hybrid automation competencies into repeatable workforce capability. By focusing on the Tier 4 hybrid skills gap, this report provides CNSI with a structured pathway to prepare automation-literate scientists who can manage AI-integrated workflows, understand cross-platform automation logic, and operate effectively in high-throughput biofoundry environments. In this sense, the final deliverable is not only a curriculum

proposal but also a workforce training blueprint to make ExFAB's educational mission more scalable, operationally feasible, and aligned with its role as a research-first biofoundry.

II. Feasibility Within Constraints

A central strength of the proposed model is that it is designed around ExFAB's operating constraints rather than assuming additional resources are available. Because ExFAB remains a research-first facility, the training model cannot depend on new full-time hires, major equipment purchases, or lab space modifications. The proposed 40-person blended bootcamp responds directly to this constraint by using a dry-studio / wet-lab rotation structure that expands training capacity without overcrowding physical lab space or disrupting ongoing research operations.

This blended model allows ExFAB to preserve its 85–90% research-first mission while still scaling its impact on workforce development. In the wet lab, trainees focus on physical competencies that require direct interaction with equipment, such as diagnosing hardware failures, tuning liquid classes, calibrating physical decks, and observing pre-validated protocol runs. In the dry studio, trainees use floating software licenses, digital twin tools, Python logic, and LIMS workflows to design and virtually execute protocols before deploying them in the wet lab. This approach reduces pressure on limited instruments while ensuring that physical equipment time is reserved for competencies that cannot be effectively simulated. As a result, the model demonstrates that ExFAB can scale training capacity, protect research operations, and remain within a Zero-FTE and No-CapEx implementation framework.

III. Economic & Operational Impact

The evidence gathered through interviews and a literature review indicates that investing in workforce upskilling is not merely an educational initiative; it is also a strategic economic strategy for the biotechnology sector and a significant revenue opportunity for CNSI. The financial model projects a net annual contribution of \$395,000 beginning in Year 2, with Year 1 remaining net positive at \$308,500. This means the program recoups its \$86,450 one-time setup investment within its first cohort cycle. Beyond the direct financial return, the operational impact extends further by grounding the curriculum in vendor-agnostic automation logic rather than proprietary, platform-specific workflows. ExFAB positions itself to set a sector standard for hybrid competency development, one that strengthens its ability to service diverse industry and academic partners over time. The adoption of digital twin-based training further reduces consumable waste and also protects high-value instrumentation from premature or undertrained operation. Hence, delivering an asset protection benefit that is compounded across every cohort. Collectively, these dynamics reframe the

proposed bootcamp as a high-leverage, low-risk infrastructure investment that aligns workforce development with CNSI's broader mission of building open, scalable, and impactful scientific infrastructure.

IV. Call to Action

As biotechnology continues its rapid transition towards AI-enabled experimentation, the demand for hybrid scientists or professionals who can translate biological intent into machine-executable workflows, structure data according to FAIR principles, and diagnose failures across biological, hardware, and software domains intensifies. As AI-enabled experimentation becomes the operational standard across biopharma, agtech, synthetic biology, and clinical diagnostics, it continues to rise. The window to establish early leadership in this space is open, and ExFAB is uniquely positioned to seize it. The recommendation is to launch a 1-2 week pilot bootcamp using real-world laboratory anomalies and ExFAB's existing instrument suite to pressure-test curriculum components, validate the blended rotation logistics, and generate the outcome data needed to refine and scale the model. This pilot, containing industry interview data, is the first operational step towards formalizing a training infrastructure that supports CNSI's long-term automation strategy, generates sustainable revenue, and positions ExFAB as a leader in shaping a workforce capable of meeting the scientific challenges and opportunities ahead.

Appendix

20260313-001. CNSI Interview. March 13th, 2026.

20260309-001. CNSI Interview. March 9th, 2026.

Delebecque, C. J., & Philp, J. C. (2019). Leveraging the biofoundry to support the bioeconomy. *Microbial Biotechnology*, 12(5), 813–816.

Mey, F., Clauwaert, J., Stock, M., Waegeman, W., & De Backer, M. (2021). Improving biofoundry operations through digital transformation. *Trends in Biotechnology*, 39(11), 1101–1111.

Stephenson, N., Land, H., & Scrutton, N. S. (2023). Biofoundries provide a ready path from synthetic biology research to the clinic. *Nature Communications*, 14, 3040.